

Part VI — Heat, Temperature, and Thermodynamics · Chapter 29

Temperature and Thermal Energy

The microscopic origin of heat, temperature, and why they are not the same thing

A cup of boiling water and a bathtub of warm water have very different temperatures — yet the bathtub contains far more thermal energy. A blacksmith plunges a glowing piece of iron into a bucket of water; the iron is hotter, but the water has thousands of times more total thermal energy. These contrasts reveal one of the most fundamental and frequently confused distinctions in all of physics: temperature and thermal energy are related, but they are not the same thing. Temperature measures something about how energy is distributed among particles. Thermal energy measures how much of it there is in total. Understanding the difference — and understanding both in terms of the motion of atoms and molecules — is the foundation of thermodynamics.

29.1 What Temperature Measures

Temperature is a measure of the average kinetic energy of the particles (atoms, molecules, or ions) in a substance. The word ‘average’ is critical: in any real substance at any temperature above absolute zero, individual particles are moving at a wide range of speeds. Some are moving fast; some are moving slowly. Temperature describes the average energy of this distribution, not any individual particle’s energy.

When you measure the temperature of a cup of coffee with a thermometer, you are ultimately measuring something about the average kinetic energy of the water molecules in that cup. A higher reading means the average water molecule is moving faster and has more kinetic energy. A lower reading means a lower average kinetic energy.

This microscopic definition of temperature — average particle kinetic energy — is one of the greatest insights of 19th-century physics. It connects the macroscopic, everyday concept of temperature (what a thermometer reads) to the microscopic, mechanical concept of energy (what particles have). It unifies the physics of motion with the physics of heat, turning thermodynamics into a branch of mechanics applied to enormous numbers of particles.

Key Point: Temperature measures how concentrated or spread out the thermal

energy is across particles — not how much there is in total. A hot pin and a warm bathtub have very different amounts of thermal energy, but their temperatures tell you about the average energy per particle in each.

29.2 Microscopic View of Matter and Particle Motion

All matter is composed of atoms and molecules in constant, random motion. Even in a solid, which appears stationary from the outside, every atom vibrates continuously around its equilibrium position in the crystal lattice. In a liquid, molecules move more freely, translating past each other while remaining loosely associated. In a gas, molecules move rapidly and freely, colliding with each other and with container walls, spending most of their time far from any other molecule.

This constant, random motion of particles is thermal motion. It is not directed motion — there is no preferred direction, and the motions of different particles are essentially independent and uncorrelated. The distribution of particle speeds in a gas at a given temperature follows the Maxwell-Boltzmann distribution: most particles have speeds near the average, but significant fractions move much faster or slower. The tail of this distribution — the fastest molecules — is responsible for many important phenomena: evaporation of liquids at temperatures below boiling, the escape of gases from planetary atmospheres, and the ignition of chemical reactions.

This microscopic picture has profound implications. It means that 'heat' in its most fundamental sense is mechanical energy — the kinetic energy of particles. The transfer of heat from one object to another is the transfer of this mechanical energy through collisions between particles. Thermodynamics is, at its deepest level, statistical mechanics: the application of Newton's laws to systems of 10^{23} particles at once.

Historical Note: The kinetic theory of gases was developed in the mid-19th century by Rudolf Clausius, James Clerk Maxwell, and Ludwig Boltzmann. Boltzmann's statistical mechanical interpretation of thermodynamics was initially rejected by many physicists who doubted the existence of atoms. Boltzmann died in 1906, just before Einstein's 1905 paper on Brownian motion provided definitive experimental proof that atoms are real.

29.3 Kinetic Theory of Temperature

The kinetic theory of gases provides a precise, quantitative connection between temperature and particle motion. For an ideal gas (a theoretical model in which particles have negligible volume and interact only through perfectly elastic collisions),

the average translational kinetic energy per particle is:

$$KE_{\text{avg}} = \frac{1}{2} m v^2_{\text{avg}} = \frac{3}{2} k_B T$$

Average translational kinetic energy per particle in an ideal gas. $k_B = 1.381 \times 10^{-23}$ J/K (Boltzmann constant). T = absolute temperature in kelvin.

This equation is remarkable. It says that the average kinetic energy of a gas particle is determined by one thing only: the absolute temperature T . It does not depend on the mass of the particle or the type of gas. A helium atom and a carbon dioxide molecule at the same temperature have the same average translational kinetic energy. Since CO_2 has more mass, it must be moving more slowly: $v_{\text{avg}} = \sqrt{(3k_B T/m)}$, so heavier molecules move slower at the same temperature.

The Boltzmann constant $k_B = 1.381 \times 10^{-23}$ J/K is the fundamental constant of thermodynamics. It converts between temperature (in kelvins) and energy (in joules), bridging the macroscopic world of heat and the microscopic world of molecules. The closely related gas constant $R = N_A k_B = 8.314$ J/(mol·K), where N_A is Avogadro's number, is the per-mole version of the same conversion.

Example — Average kinetic energy at room temperature At $T = 293$ K (about 20°C): $KE_{\text{avg}} = (3/2)(1.381 \times 10^{-23})(293) = 6.07 \times 10^{-21}$ J. This is tiny on human scales but enormous relative to chemical bond energies ($\sim 10^{-19}$ J), which is why thermal energy drives chemistry.

29.4 Thermal Energy vs. Temperature

The single most important distinction in this chapter: thermal energy is not the same as temperature. They are related but fundamentally different concepts, and confusing them leads to persistent errors in reasoning about heat and thermodynamics.

Temperature is an intensive property — it describes something about each particle independently of how many particles there are. If you take a cup of hot water and split it into two cups, each cup has exactly the same temperature as the original. Temperature did not divide; it is the same in each half.

Thermal energy is an extensive property — it depends on how much material there is. Split the cup into two cups and each cup has half the thermal energy of the original. Thermal energy is proportional to the number of particles: doubling the amount of a substance at the same temperature doubles its thermal energy.

Property	Temperature	Thermal Energy
What it measures	Average KE per particle	Total KE of all particles

Type	Intensive (per particle)	Extensive (depends on amount)
Depends on quantity?	No	Yes (proportional to N or mass)
Example: split a sample	Each half has same temperature	Each half has half the energy
Unit	K, °C, °F	J (joules)

Table 29.1 *Key differences between temperature and thermal energy.*

Example: A lit match and a roaring bonfire. The match flame has a higher temperature (around 1500°C) than the outer edge of the bonfire (around 600°C). But the bonfire contains vastly more thermal energy because it involves the combustion of millions of times more material. The bonfire can heat a room; the match cannot. Thermal energy, not temperature, determines how much heating a source can do.

29.5 Internal Energy

The internal energy of a system (symbol U) is the total energy stored in the microscopic degrees of freedom of the system — the kinetic energy of all the particles plus the potential energy of their interactions (chemical bonds, intermolecular forces). In an ideal monatomic gas, the internal energy is purely kinetic and equals $(3/2)Nk_B T$, where N is the number of particles.

For real substances, internal energy includes more than just translational kinetic energy. Diatomic and polyatomic molecules can rotate and vibrate, storing energy in these additional modes. Liquids and solids have significant potential energy stored in the interactions between molecules. Phase changes involve large changes in internal energy (potential energy of bonding) with no change in temperature.

The distinction between internal energy and thermal energy is subtle at this level but matters in advanced work. Loosely, thermal energy refers to the energy associated with random thermal motion (what can be transferred as heat), while internal energy also includes chemical and nuclear energy that is not directly available for heat transfer. For the purposes of this chapter and the next two, you may use the terms interchangeably with care.

The change in internal energy of a system is central to the First Law of Thermodynamics (Chapter 32): $\Delta U = Q - W$, where Q is heat added to the system and W is work done by the system. Understanding what U is physically — the microscopic

mechanical energy of the system's particles — is the foundation for understanding why this law holds.

29.6 Thermal Equilibrium

Thermal equilibrium is the state in which two objects in thermal contact have reached the same temperature and no further net heat flow occurs between them. When objects at different temperatures are brought into contact, the hotter object transfers energy to the cooler object until both reach a common temperature — the equilibrium temperature.

Thermal equilibrium is governed by the Zeroth Law of Thermodynamics, which is so fundamental that it was named 'zeroth' because it was recognized after the First and Second Laws had already been numbered, yet logically precedes them: If system A is in thermal equilibrium with system C, and system B is also in thermal equilibrium with system C, then A and B are in thermal equilibrium with each other.

The Zeroth Law justifies the concept of temperature as a well-defined quantity. It means that temperature is transitive: objects at the same temperature are in mutual equilibrium. This is what makes thermometers work — a thermometer in equilibrium with an object has the same temperature as that object.

The approach to equilibrium is always from the warmer toward the cooler. Heat flows spontaneously from high temperature to low temperature, never in reverse. This is the basis of the Second Law of Thermodynamics, which will be developed in Chapter 33.

The Rule: Thermal equilibrium: $T_A = T_B \rightarrow$ no net heat flow. When two objects at different temperatures are in contact, heat flows from the higher-temperature object to the lower-temperature object until they reach a common equilibrium temperature.

29.7 Heat vs. Temperature

Heat (symbol Q) is thermal energy in transit — energy that flows from one system to another because of a temperature difference. Heat is not a property that a system possesses (like temperature or internal energy); it is an energy transfer process. Once transferred, the energy becomes part of the recipient's internal energy; it is no longer 'heat' in the physical sense.

The distinction: a hot iron contains internal energy; it does not 'contain heat' in the way it contains mass. When the hot iron is placed on a cool ironing board, energy flows as heat from the iron to the board. After equilibrium is reached, neither object

contains heat — both have internal energy at the same temperature. Heat existed only during the transfer process.

This may seem like pedantry, but it has concrete consequences. You cannot say “an object has X joules of heat.” You can say “an object’s internal energy increased by X joules due to heat transfer.” The distinction becomes essential when analyzing thermodynamic processes in which both heat and work transfer energy simultaneously.

Common Mistake: Heat is not a substance. Saying “there is more heat in boiling water than in ice” confuses heat (a process) with internal energy (a property). The correct statement is: boiling water has higher internal energy per unit mass than ice at the same mass. Heat is the transfer process by which the difference is established.

29.8 Temperature Scales

Three temperature scales are in common use: Celsius, Fahrenheit, and Kelvin.

The Celsius scale (symbol $^{\circ}\text{C}$) defines 0°C as the freezing point of water at standard atmospheric pressure and 100°C as the boiling point. It is the scale used in everyday life outside the United States and in most scientific contexts outside of thermodynamics.

The Fahrenheit scale (symbol $^{\circ}\text{F}$) defines 32°F as the freezing point of water and 212°F as the boiling point. It is in common use in the United States for everyday temperature. The degree Fahrenheit is $5/9$ the size of a degree Celsius.

The Kelvin scale (symbol K, no degree symbol) is the SI unit of thermodynamic temperature. Its zero point is absolute zero — the temperature at which all thermal motion theoretically stops. The size of one kelvin equals the size of one degree Celsius (the scales have the same step size). Therefore, a temperature difference of 10 K is the same as a temperature difference of 10°C .

Reference Point	Kelvin (K)	Celsius ($^{\circ}\text{C}$)	Fahrenheit ($^{\circ}\text{F}$)
Absolute zero	0	-273.15	-459.67
Freezing point of water	273.15	0	32
Body temperature	310.15	37	98.6
Boiling point of water	373.15	100	212
Surface of the Sun	≈ 5800	≈ 5527	≈ 9980

Table 29.2 Common reference temperatures on three scales.

Why use Kelvin for physics? Because the kinetic theory equation $KE = (3/2)k_B T$ only works if T is in kelvins. Celsius and Fahrenheit have arbitrary zero points that do not correspond to zero particle kinetic energy. Kelvin is the only scale whose zero corresponds to zero thermal energy, making it the natural scale for thermodynamic equations.

29.9 Converting Between Temperature Scales

The three conversion formulas between temperature scales are:

$$K = ^\circ C + 273.15$$

Celsius to Kelvin. Add 273.15 (often approximated as 273).

$$^\circ C = (5/9) (^\circ F - 32)$$

Fahrenheit to Celsius. Subtract 32, then multiply by 5/9.

$$^\circ F = (9/5) ^\circ C + 32$$

Celsius to Fahrenheit. Multiply by 9/5, then add 32.

Step 1: F to K: First convert $F \rightarrow C$, then $C \rightarrow K$. Two steps.

Step 2: Note: Temperature differences in K equal temperature differences in $^\circ C$: $\Delta T_K = \Delta T_{^\circ C}$.

Step 3: Caution: Temperature differences in Fahrenheit need the 5/9 factor: $\Delta T_{^\circ C} = (5/9)\Delta T_{^\circ F}$.

Example — Convert body temperature $37.0^\circ C$ to Kelvin: $K = 37.0 + 273.15 = 310.15$ K. To Fahrenheit: $^\circ F = (9/5)(37.0) + 32 = 66.6 + 32 = 98.6^\circ F$.

Example — Convert a temperature difference A furnace temperature rises by $450^\circ F$. In $^\circ C$: $\Delta T = (5/9)(450) = 250^\circ C = 250$ K. A difference of $450^\circ F$ equals a difference of $250^\circ C$, not $450^\circ C$.

29.10 Absolute Zero

Absolute zero is the lowest possible temperature: $0\text{ K} = -273.15^\circ C$. At absolute zero, a classical ideal gas would have zero pressure and zero volume; quantum mechanically, all systems would be in their lowest possible energy state (though quantum mechanics requires a small residual 'zero-point energy' even at 0 K).

Absolute zero is a theoretical limit that cannot be reached in practice, though scientists have cooled substances to within billionths of a kelvin of it. The Third Law of Thermodynamics states that no finite sequence of physical processes can reduce the temperature of a system to absolute zero in a finite number of steps.

The concept of absolute zero gives physical meaning to temperature. Negative temperatures in Kelvin are impossible (for normal systems), because you cannot have less than zero thermal kinetic energy. The Kelvin scale is therefore bounded below by zero, giving it a physically motivated origin unlike Celsius or Fahrenheit.

At extremely low temperatures, materials exhibit quantum behavior invisible at room temperature. Superconductivity (zero electrical resistance) occurs in many metals below 10–20 K. Superfluidity (zero-viscosity flow) occurs in liquid helium below 2.17 K. Bose-Einstein condensates — a new state of matter in which many atoms occupy the same quantum state — form at temperatures of 100 nanokelvin or less. These phenomena are entirely outside classical physics.

Historical Note: Lord Kelvin (William Thomson, 1824–1907) proposed the absolute temperature scale in 1848, motivated by the observation that an ideal gas's pressure decreases linearly with temperature and would reach zero at -273°C . He recognized this as a natural zero point for temperature. The SI unit of temperature, the kelvin, is named in his honor.

29.11 Thermal Expansion

Almost all materials expand when heated and contract when cooled. This thermal expansion occurs because increasing temperature increases the average kinetic energy of particles, which causes them to vibrate more energetically. In solids and liquids, this increased vibration amplitude pushes particles slightly farther apart on average, increasing the material's dimensions.

Linear thermal expansion (change in length): for a solid rod of length L_0 at temperature T_0 , when the temperature changes by ΔT , the change in length is:

$$\Delta L = \alpha L_0 \Delta T$$

Linear thermal expansion. α = coefficient of linear expansion ($1/^{\circ}\text{C}$ or $1/\text{K}$); L_0 = original length; ΔT = temperature change. Units: ΔL in same units as L_0 .

Volume thermal expansion (change in volume): for a solid or liquid of volume V_0 :

$$\Delta V = \beta V_0 \Delta T$$

Volume thermal expansion. β = coefficient of volume expansion $\approx 3\alpha$ for solids (expansion occurs in all three dimensions). Liquids and gases have tabulated β values directly.

Material	Linear ($10^{-6}/^{\circ}\text{C}$)	Coeff. α	Volume ($10^{-6}/^{\circ}\text{C}$)	Coeff. β
Aluminium	23		69	
Brass	19		57	
Steel	11		33	
Glass (ordinary)	9		27	
Invar (Fe-Ni alloy)	1.2		3.6	
Concrete	12		36	
Water (at 20°C)	—		207	
Mercury	—		182	
Air (at 20°C)	—		≈ 3400	

Table 29.3 Thermal expansion coefficients at approximately 20°C .

Example — Bridge expansion A steel bridge span is 200 m long at 10°C . At 40°C ($\Delta T = 30^{\circ}\text{C}$): $\Delta L = 11 \times 10^{-6} \times 200 \times 30 = 0.066 \text{ m} = 6.6 \text{ cm}$. Expansion joints must accommodate this change to prevent buckling.

29.12 Real-World Effects of Thermal Expansion

Thermal expansion has significant engineering and practical consequences across many scales.

Expansion joints and gaps are deliberately built into bridges, railway tracks, pipelines, and concrete pavements to allow thermal expansion without causing structural damage. Railway tracks laid without gaps have buckled on hot summer days, and pipelines without expansion loops have ruptured in cold winters. The Eiffel Tower is about 15 cm taller in summer than in winter due to thermal expansion of its iron structure.

Bimetallic strips exploit the difference in thermal expansion coefficients between two metals bonded together. When temperature changes, the strip bends because one metal expands more than the other. This bending is used in thermostats, circuit breakers, and mechanical thermometers. A bimetallic thermostat strip bends to open or close an electrical circuit at a preset temperature.

Water has an anomalous expansion behavior near its freezing point: unlike most substances, water is denser at 4°C than at 0°C (ice). As water cools below 4°C, it expands rather than contracts, which is why ice floats on water and lakes freeze from the top rather than from the bottom. This anomaly is critical to aquatic life — bottom-dwelling organisms survive winter because liquid water remains beneath the ice layer.

Thermal expansion also matters in precision instruments. The pendulum of a mechanical clock must maintain a constant length for constant timekeeping; temperature changes cause pendulum rods to expand, slowing the clock. Early clockmakers solved this using temperature-compensated pendulums with different metals arranged so their expansion effects cancel. Modern atomic clocks and GPS receivers must account for thermal effects in their electronic components at parts-per-billion levels.

Study Tip: When solving thermal expansion problems: $\Delta L = \alpha L_0 \Delta T$. Always use the original length L_0 (not the final length). ΔT can be in °C or K (they are equal in magnitude). Include sign: expansion for positive ΔT , contraction for negative ΔT .

29.13 Common Mistakes in Thermal Physics

- Confusing temperature and heat. Temperature is a property of a system (average particle KE). Heat is energy in transit due to temperature difference. A system has a temperature; heat flows between systems. Never say “a substance contains heat.”
- Confusing temperature and thermal energy. A small hot object can have higher temperature but far less total thermal energy than a large cool object. Temperature is intensive; thermal energy is extensive.
- Using Celsius instead of Kelvin in kinetic theory equations. The relationship $KE = (3/2)k_B T$ requires T in kelvins. Substituting Celsius values gives completely wrong results.
- Forgetting that temperature differences in Kelvin and Celsius are the same. $\Delta T = 30\text{ K} = 30^\circ\text{C}$. You do not need to add 273 when working with differences, only with absolute temperatures.
- Applying the conversion $T_K = T_C + 273$ to temperature differences. If a problem says a temperature rises by 50°C , the change is 50 K — you do not add 273 again.
- Misidentifying the original length in thermal expansion. $\Delta L = \alpha L_0 \Delta T$ uses the length before the temperature change, L_0 , not the final length. The error is small for small ΔT but matters in precise calculations.

- Treating thermal expansion as proportional to temperature rather than to temperature change. ΔL depends on ΔT , not on the absolute temperature T .
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Chapter Summary

Temperature is a measure of the average kinetic energy of particles in a substance. Thermal energy is the total kinetic energy of all particles. They are not the same: temperature is intensive, thermal energy is extensive.

- Kinetic theory: $KE_{avg} = (3/2)k_B T$. All molecules at the same temperature have the same average translational KE regardless of mass. Heavier molecules move slower.
 - Internal energy U = total microscopic energy (kinetic + potential) of a system.
 - Thermal equilibrium: no net heat flow between systems at the same temperature. Zeroth Law: if A and C are in equilibrium, and B and C are in equilibrium, then A and B are in equilibrium.
 - Heat Q is energy in transit due to temperature difference — not a property stored in objects.
 - Temperature scales: $K = ^\circ C + 273.15$. $^\circ C = (5/9)(^\circ F - 32)$. Kelvin is required for thermodynamic equations.
 - Absolute zero ($0\text{ K} = -273.15^\circ C$) is the theoretical minimum temperature, approached but never reached.
 - Thermal expansion: $\Delta L = \alpha L_0 \Delta T$ (linear); $\Delta V = \beta V_0 \Delta T$ (volume). Water anomalously expands below $4^\circ C$.
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Key Terms

temperature: A measure of the average translational kinetic energy of the particles in a substance; an intensive property; measured in K, $^\circ C$, or $^\circ F$.

thermal energy: The total kinetic energy of all particles in a substance due to random thermal motion; an extensive property proportional to the number of particles and to temperature.

internal energy (U): The total microscopic energy of a system, including kinetic energy of particles and potential energy of their interactions.

kinetic theory: The model that describes the macroscopic properties of matter (temperature, pressure) in terms of the microscopic motion of atoms and molecules.

Boltzmann constant (k_B): $k_B = 1.381 \times 10^{-23} \text{ J/K}$; the fundamental constant relating temperature to particle kinetic energy: $KE_{\text{avg}} = (3/2)k_B T$.

thermal equilibrium: The state in which two systems in thermal contact reach the same temperature and no net heat flow occurs between them.

Zeroth Law of Thermodynamics: If A is in thermal equilibrium with C, and B is in thermal equilibrium with C, then A and B are in thermal equilibrium with each other; justifies the concept of temperature.

heat (Q): Thermal energy in transit: energy transferred from one system to another because of a temperature difference. Not a property stored in a system.

Celsius scale ($^{\circ}\text{C}$): Temperature scale with 0°C at water's freezing point and 100°C at its boiling point at standard atmospheric pressure.

Kelvin scale (K): The SI thermodynamic temperature scale with 0 K at absolute zero and the same degree size as Celsius; $K = ^{\circ}\text{C} + 273.15$.

absolute zero: The lowest possible temperature: $0 \text{ K} = -273.15^{\circ}\text{C}$; the theoretical point of minimum thermal energy.

thermal expansion: The increase in the dimensions of a material when its temperature rises; described by $\Delta L = \alpha L_0 \Delta T$ for linear expansion.

coefficient of linear expansion (α): Material property describing how much length increases per unit length per degree of temperature rise; units of $^{\circ}\text{C}^{-1}$ or K^{-1} .

bimetallic strip: A device consisting of two bonded metals with different expansion coefficients that bends when temperature changes; used in thermostats and circuit breakers.

Maxwell-Boltzmann distribution: The statistical distribution of molecular speeds in a gas at a given temperature; shows most molecules near an average speed with a tail of faster molecules.

Concept Review Questions

Answer in complete sentences. No calculations required unless stated.

1. Explain the difference between temperature and thermal energy. Why does a bathtub of warm water contain more thermal energy than a cup of boiling water?

2. According to kinetic theory, what happens at the molecular level when you heat a gas? How does this change its temperature?
 3. A helium atom and an oxygen molecule (O_2) are at the same temperature. (a) Which has greater average kinetic energy? (b) Which is moving faster on average? Explain.
 4. Explain what heat is. Why is it incorrect to say that an object 'contains heat'?
 5. Why must temperature be expressed in kelvins when used in the kinetic theory equation $\text{KE} = (3/2)k_B T$? What goes wrong if you use Celsius?
 6. What is the Zeroth Law of Thermodynamics? Explain how it justifies the use of a thermometer to measure temperature.
 7. Water reaches its maximum density at 4°C rather than at 0°C . Explain why this anomaly is critical for the survival of aquatic life in winter.
 8. A bimetallic strip curves toward the metal with the smaller expansion coefficient when heated. Explain why, using the concept of differential thermal expansion.
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Temperature Conversion Problems

9. Convert the following to Kelvin: (a) 25°C , (b) -40°C , (c) 1500°C (surface of tungsten filament), (d) -196°C (boiling nitrogen).
 10. Convert the following to Celsius: (a) 300 K, (b) 0 K, (c) 5800 K (surface of Sun), (d) 77 K.
 11. Convert the following to Fahrenheit: (a) 100°C , (b) -40°C , (c) 37°C , (d) 20°C .
 12. A fever is reported as 103.5°F . Convert to Celsius and Kelvin.
 13. An oven is heated from 150°C to 250°C . (a) What is the change in Celsius? (b) What is the change in Kelvin? (c) What is the change in Fahrenheit?
 14. Liquid nitrogen boils at 77 K. (a) What is this in Celsius? (b) In Fahrenheit? (c) If a sample warms from 77 K to 293 K, what is the temperature rise in $^\circ\text{C}$ and in $^\circ\text{F}$?
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Thermal Expansion Problems

15. A steel railway rail is 18.0 m long at 5°C . What is its length at 40°C ? ($\alpha_{\text{steel}} = 11 \times 10^{-6} / ^\circ\text{C}$)

16. A brass rod has length 1.500 m at 20°C. Find its length at (a) 80°C and (b) –10°C. ($\alpha_{\text{brass}} = 19 \times 10^{-6} / ^\circ\text{C}$)
 17. The Eiffel Tower is 330 m tall. Its iron structure has $\alpha = 12 \times 10^{-6} / ^\circ\text{C}$. If the temperature varies between –10°C in winter and 35°C in summer, by how much does the tower's height change?
 18. A glass sphere has volume 500 cm³ at 20°C. Find its volume at 80°C. ($\beta_{\text{glass}} = 27 \times 10^{-6} / ^\circ\text{C}$)
 19. An aluminium ring at 20°C has an inner diameter of exactly 5.000 cm. A steel shaft at 20°C has a diameter of 5.010 cm. To what temperature must the ring be heated to just slip over the shaft? ($\alpha_{\text{Al}} = 23 \times 10^{-6} / ^\circ\text{C}$; steel shaft is not heated.)
 20. A pendulum clock uses an invar rod ($\alpha = 1.2 \times 10^{-6} / ^\circ\text{C}$) of length 1.000 m at 20°C. How much does the rod length change between summer (35°C) and winter (5°C)? Compare this to a steel pendulum ($\alpha = 11 \times 10^{-6} / ^\circ\text{C}$).
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Thermal Energy Analysis

21. At 300 K, calculate the average translational kinetic energy of a gas molecule. ($k_B = 1.381 \times 10^{-23} \text{ J/K}$)
 22. At what temperature does the average translational KE of a molecule equal 1.0 eV? (1 eV = $1.60 \times 10^{-19} \text{ J}$)
 23. A nitrogen molecule (N₂, mass = $4.65 \times 10^{-26} \text{ kg}$) is in air at 20°C. Find its average translational kinetic energy and estimate its root-mean-square speed $v_{\text{rms}} = \sqrt{3k_B T/m}$.
 24. Explain why a large lake takes much longer to heat up or cool down than a small puddle, even though both are at the same temperature initially. Use the distinction between thermal energy and temperature in your answer.
 25. A 500 g iron block and a 500 g aluminium block are both at 80°C. Do they have the same thermal energy? Explain. (Hint: think about what determines the thermal energy per unit mass.)
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Applied Thermal Physics Exercises

26. A steel pipe is 50.0 m long at 20°C. It carries steam at 200°C. (a) How much does the pipe expand? (b) Design an expansion loop: the loop must accommodate this change. If the loop is U-shaped with two arms of length x , how long must x be if the loop material can deflect by 5 mm per meter of arm length?
27. A bimetallic thermostat strip consists of brass (outer, $\alpha = 19 \times 10^{-6} / ^\circ\text{C}$) bonded to invar (inner, $\alpha = 1.2 \times 10^{-6} / ^\circ\text{C}$). When the temperature rises, which direction does the strip curve (toward brass or toward invar side)? Explain physically.
28. An ideal gas thermometer contains gas at volume $V_0 = 1.000 \text{ L}$ at 273.15 K. If the gas pressure increases by 10%, what is the new temperature? (For constant volume ideal gas: $P/T = \text{constant}$.)
29. The average speed of air molecules at 20°C is about 500 m/s. (a) Calculate the kinetic energy of one nitrogen molecule at this speed ($m = 4.65 \times 10^{-26} \text{ kg}$). (b) The speed of sound in air is 343 m/s. Explain why the sound speed is related to but not equal to the molecular speed.

Chapter 29 Checkpoint

Before moving on to Chapter 30, confirm you can do each of the following.

I can...	See Section
Define temperature as a measure of average particle kinetic energy	29.1
Describe thermal motion in solids, liquids, and gases at the microscopic level	29.2
State and apply $\text{KE}_{\text{avg}} = (3/2)k_B T$; use SI units consistently	29.3
Distinguish temperature (intensive) from thermal energy (extensive)	29.4
Define internal energy and explain what it includes beyond translational KE	29.5
Define thermal equilibrium and state the Zeroth Law of Thermodynamics	29.6
Explain what heat is and why "a substance contains heat" is	29.7

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Compare Celsius, Fahrenheit, and Kelvin scales; state why K is required in equations	29.8
Convert temperatures between K, °C, and °F; handle differences correctly	29.9
Define absolute zero and state the Third Law implication	29.10
Apply $\Delta L = \alpha L_0 \Delta T$ and $\Delta V = \beta V_0 \Delta T$ for thermal expansion	29.11
Explain three engineering examples of thermal expansion effects	29.12
Identify and correct common errors in temperature and thermal energy problems	29.13

Continue to Chapter 30: Heat Transfer →